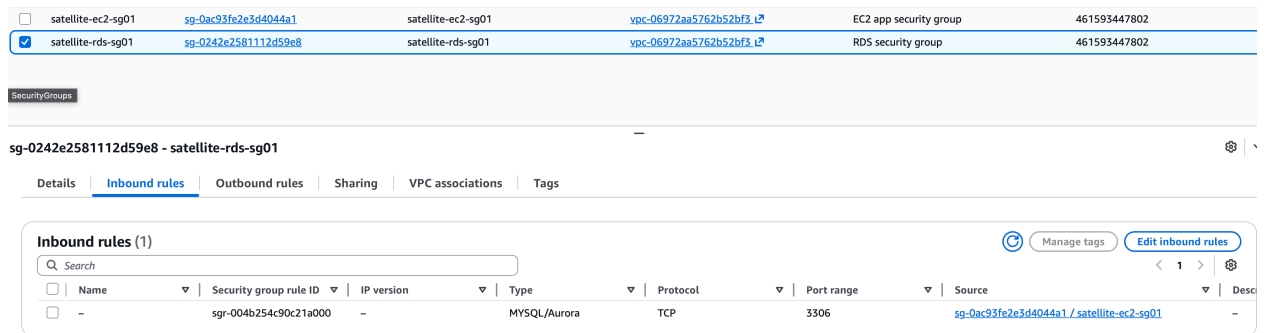


Kevin Willocks Lab 1 A submission:

Endpoint: satellite-rds01.cef7de7jol4f.us-east-1.rds.amazonaws.com

1) Screenshots:

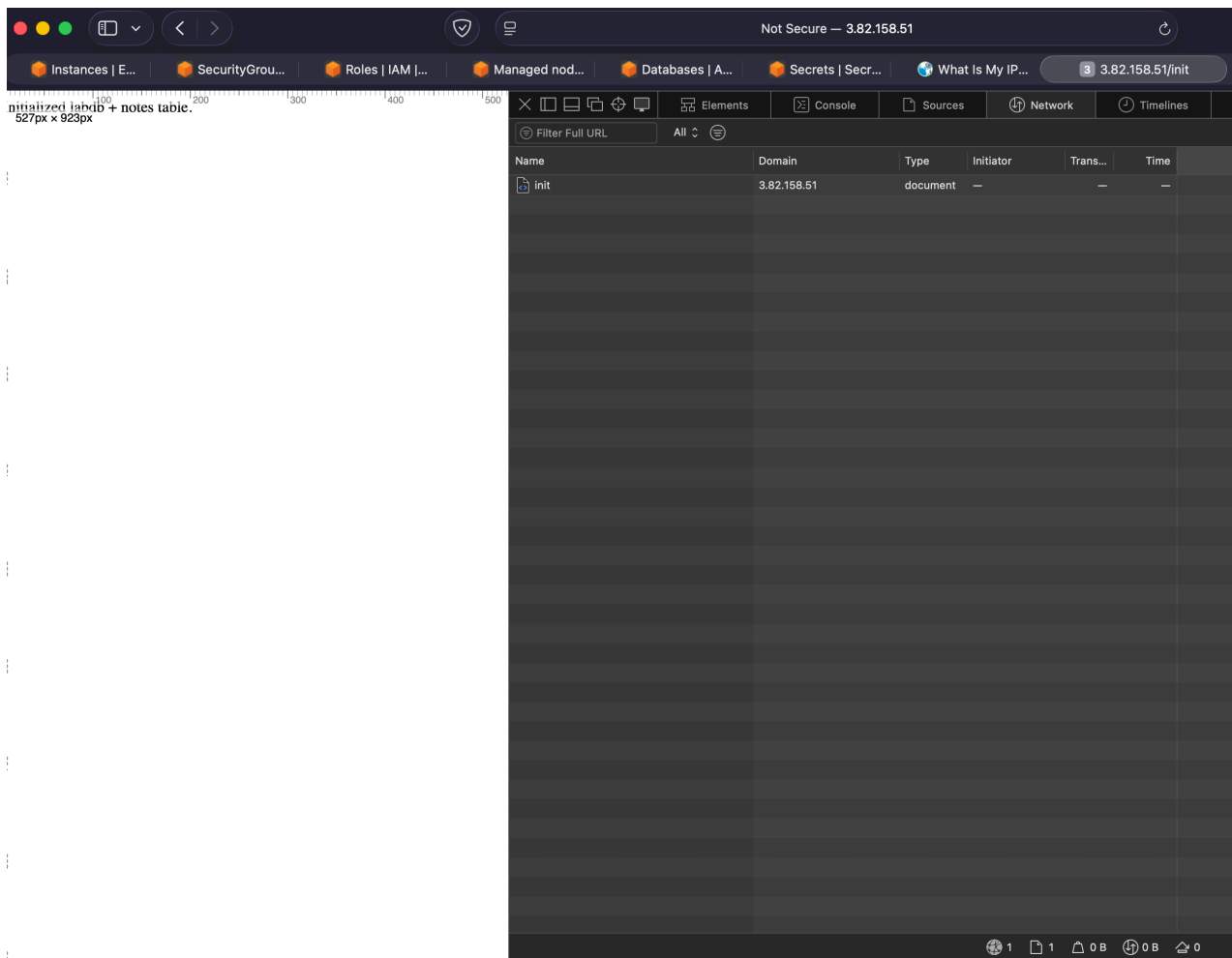
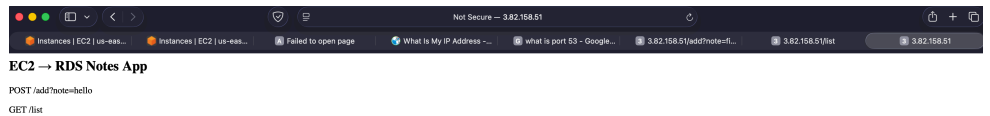
A) RDS SG inbound rule:



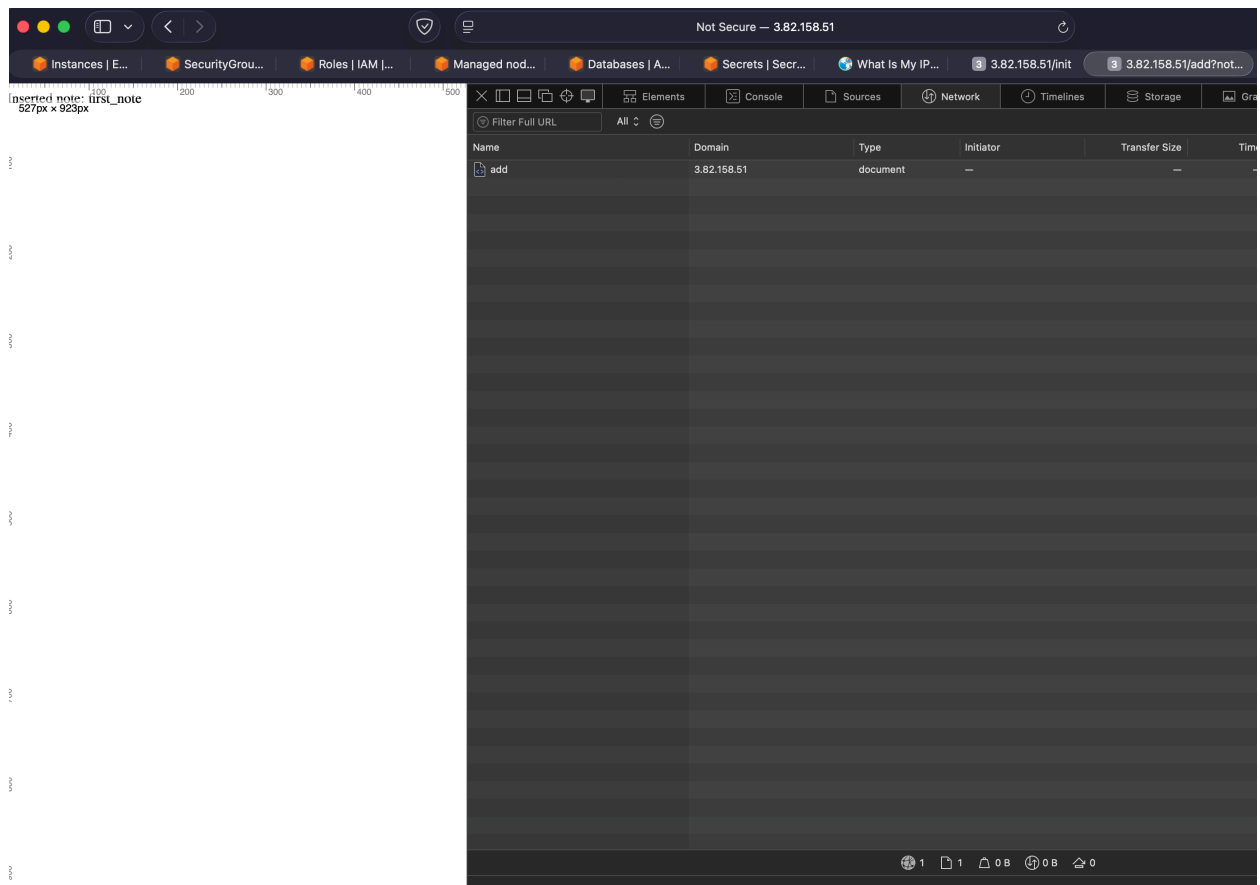
The screenshot shows the AWS IAM console's 'Inbound rules' tab for the security group 'sg-0242e2581112d59e8' (satellite-rds-sg01). The table below lists the inbound rules.

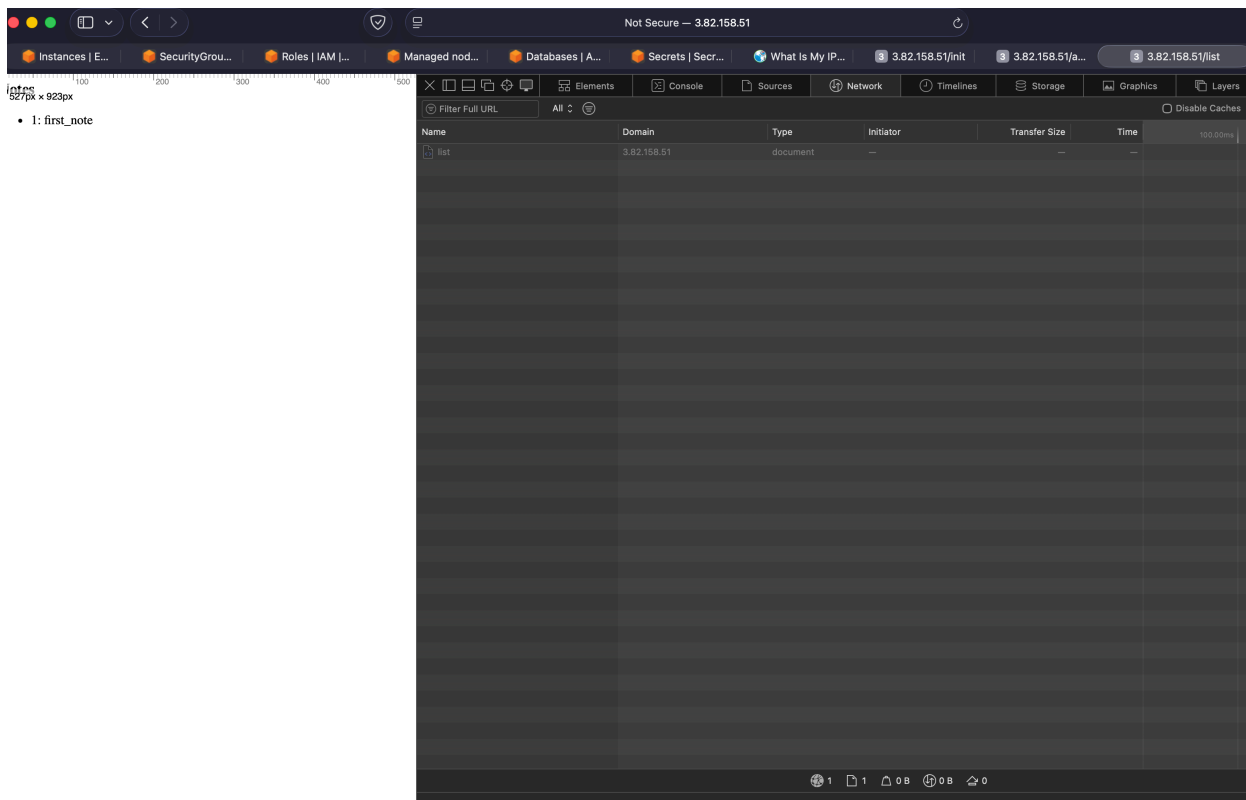
Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sg-004b254c90c21a000	-	MySQL/Aurora	TCP	3306	sg-0ac93fe2e3d4044a1 / satellite-ec2-sg01

B) <http://3.82.158.51/init>



C) http://3.82.158.51/add?note=first_note





C) <http://3.82.158.51/list>

2) The reason why DB inbound source is restricted to the EC2's security group is because, we want to control who/what has access to our DB. The DB contains sensitive data (usernames and passwords) and any breach of a company's database that contains this sensitive data will result in financial, trust and reputational losses. MySQL uses port 3306 which provides users the ability to manage DBs.

Secrets Manager is much better and preferred over hardcoding credentials due to several reasons. The first is management. When using tools such as Github, not every developer will

have an up to date branch containing the current credentials. This would cause developers to continually change the credentials after every merge even with implementing Pull Requests. The second is security. Should an unauthorized party access the DB, they will have unfettered access to all the content the DB has. Secrets Manager is an AWS managed service that provides users the ability to encrypt secrets, control access and rotate the secrets, making it much more effective than hardcoding secrets.